integrating artificial intelligence and optimization techniques into mechanical systems.

Dr. Ranjan’s research and professional profiles are available on SCOPUS, Web of Science, ORCID, Google Scholar, and Vidwan. His work bridges theoretical research with practical engineering solutions, fostering innovation in mechanical engineering and allied technologies.

Dr. Onur Mesut Şenaras is a distinguished engineer and academic specializing in automotive engineering, industrial optimization, and lean manufacturing. He earned his B.Sc. in Industrial Engineering from Dokuz Eylül University (2004), his M.Sc. in Automotive Engineering from Uludağ University (2013), and his Ph.D. in Automotive Engineering from Uludağ University (2024).

With nearly two decades of professional experience, Dr. Şenaras began his career as Controlling and Budget Responsible at Vestel Beyaz Eşya (2005–2007) and subsequently joined Oyak Renault, holding key roles such as Body Performance Responsible, Kaizen Pilot, and Design Automation Leader. His contributions have driven digitalization, automation, and system optimization within the automotive sector. He has received multiple prestigious awards, including the Renault Ergonomics Award (2010), Renault Investment Optimization Award (2011), and several factory ergonomics awards, and holds a national patent in ergonomics.

As a researcher and author, Dr. Şenaras focuses on automated guided vehicles (AGVs), artificial intelligence, system dynamics, and simulation. He co-authored *Expulsion in Resistance Spot Welding* (2017) and has contributed chapters on flexible manufacturing systems and subway passenger flow modeling. At Uludağ University, he mentors students and teaches courses on lean manufacturing, logistics management, and genetic